OSI MODEL

Layered Approach



Application Layer

The protocols apps use directly. Example: Browsing with HTTP/HTTPS, sending email via SMTP, DNS resolving "google.com" to an IP, FTP file transfers.

Presentation Layer

Encryption, compression, data formatting. TLS encrypts your online banking session; JPEG/MP3 compression; JSON/XML formatting between systems.

Session Layer

Opening, maintaining and closing conversations. Staying logged in while a video meeting reconnects after a brief drop; remote desktop session resume.

Transport Layer

End-to-end delivery (TCP reliable, UDP fast). TCP ensures a file download arrives complete; UDP is used for video calls where speed beats perfection.

Network Layer

IP addressing and routing between networks. Your router sends traffic from home to a server in another country; ping and traceroute test this layer.

Data Link Layer

MAC addressing and switching within a local network. An office switch delivers a file to the right PC by MAC address. The IP packet goes through a framing process.

Physical Layer

Cables, Wi-Fi radio signals, fiber. This is how data is physically transported.